Shengwu Zhao

Basic Information

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Date of birth: 1998.07
Birthplace: Hefei City, Anhui Province, China
Topic: Inertial navigation and underwater geophysical field matching navigation
English level: CET-4/570, CET-6/474
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Education experience:

- Doctor (Sep, 2020 - present):

School: Beijing Institute of Technology Major: Control Science and Engineering

- Undergraduate (Sep, 2016 - June, 2020):

School: Beijing Institute of Technology Major: Control Science and Engineering

Awards

Doctor:

August	2022	Technical Invention Award of Chinese Society of Inertial Technology (*First Prize*)
January	2022	APMCM Mathematical Modeling (*First Prize*)
December	2021	Huawei Cup Mathematical Modeling (*Third Prize*)
June	2021	MathorCup Mathematical Modeling (*Third Prize*)

Undergraduate:

July	2020	The Artificial Intelligence Challenge of Robomaster, perception (*Excellence Award*)
July	2020	NXP National College Student Smart Car Competition (*Second Prize in North China*)
January	2019	China Undergraduate Mathematical Contest in Modeling (*Second Prize in Beijing*)

Research experience

- **Hardware circuit design of intelligent energy-saving car (2019.9 – 2020.7)**

  In the case that the battery is not applicable as a power source, the coil and the changing magnetic field are used to provide power for the smart car, so that the car can complete the driving within the specified time. My work is to design the circuit of the whole car and the design of the PCB, including the voltage regulator circuit design, control circuit design and signal circuit design. After that, as vice captain of the team club, participated in guiding the preparation and organization of the next race.

- **Robomaster robot positioning and decision making (2019.9 – 2020.5)**

  In the specified area, four robots need to fight against enemy robots, by designing different decision-making strategies to defeat the enemy robots. I work on robot positioning and decision making. The calculation amount is reduced by optimizing the AMCL algorithm, and the comparison of several algorithms is made. Handle battlefield complexities by designing different decisions, such as how robots behave when they don't have bullets.

- **Underwater geophysical field matching navigation (2020.9 – present)**

  The accumulated error of inertial navigation system is corrected by using underwater geophysical field for matching and positioning. My work content is to study the underwater carrier path planning method and matching positioning method. In the field of path planning, I consider the change rate of gravity field and mismatching phenomenon, and use Astar and DWA algorithms to plan the trajectory of underwater vehicles. In order to integrate inertial navigation information, geophysical field reference map and gravimeter measurement information, some matching algorithms are designed to realize positioning. It includes introducing the correlation filter in target recognition and particle filter, considering the potential relationship of geophysical field characteristics.

  In the past, I worked on gravity field aided navigation. Now I work on terrain aided navigation and positioning. The localization of underwater vehicle is studied by inertial navigation system, single-beam echosounder or multi-beam echosounder.

  As the first author, 3 SCI papers have been published and 1 patent is under review. At present, two papers are under review and two are being written.

  In this process, I have mastered knowledge about inertial navigation, integrated navigation, filter algorithm. My current research direction is integrated navigation of geophysical fields, including gravity field and topographic field.

Academic achievements

[1] **Zhao S**, Xiao X, Deng Z, et al. Gravity Matching Algorithm Based on Correlation Filter[J]. IEEE Sensors Journal, 2022, 23(3): 2618-2629.

[2] **Zhao S**, Xiao X, Wang Y, et al. An Improved Particle Filter Based on Gravity Measurement Feature in Gravity-Aided Inertial Navigation System[J]. IEEE Sensors Journal, 2022, 23(2): 1423-1435.

[3] **Zhao S**, Shi L, Zhang W, et al. Global dynamic path-planning algorithm in gravity-aided inertial navigation system[J]. IET Signal Processing, 2021, 15(8): 510-520.

[4] Wang Y, Deng Z, Zhang W, **Zhao S**. An improved ICCP gravity matching algorithm based on Mahalanobis distance[C]//2021 40th Chinese Control Conference (CCC). IEEE, 2021: 3503-3508.

[5] Deng Z, **Zhao S**, Wang Y, et al. Gravity matching method based on the characteristics of gravity measurement data [P]. Beijing：CN115183770A,2022-10-14.

[6] **Zhao S**, Xiao X, Pang X, et al. Gravity Matching Algorithm Based on Backtracking for Small Range Adaptation Area. 2023. Under Review, IEEE Transactions on Instrumentation and Measurement.

[7] **Zhao S**, Deng Z, Zhang W, et al. Adaptive Point Mass Filter and its Application in Terrain Matching Navigation. 2023. Under Review.